

Curriculum Vol 0 Chapter 6 — Integer Web Principle (Chapter-Grade Draft v0.4 — Mersenne tower + three-layer over-determinism + α -analog Integer Web absorption)

Keeper (original) + Lyra (Friday v0.3→v0.4 prose depth-investment)

2026-05-21 Thursday 10:34 EDT initial; Friday 2026-05-22 ~10:33
EDT v0.4 prose absorption per Casey + Keeper textbook completion
phase

Vol 0 Chapter 6 — Integer Web Principle

Chapter motivation

In standard physics, fundamental constants are isolated numbers: $\alpha \approx 1/137$, $m_p/m_e \approx 1836.15$, $\sin^2\theta_W \approx 0.23$, etc. Each is measured experimentally; relationships between them (where they exist) emerge from physics models.

In BST, the BST primary integers (rank=2, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$, $N_{\max}=137$) are NOT isolated — each holds a web of structural relations linking it to other primaries, to physical observables, to mathematical structures across the BST catalog.

This is Casey's Integer Web Principle — a substrate-structural observation that the primaries are foundational anchors for integer-webs of substrate-significant relations.

Section 6.1 — Casey's vision of the Integer Web

Per Casey Wednesday May 20 vision (filed as Casey-named principle):

“Inside the substrate, the boundaries are the integers, each holding a web.”

Interpretation: - The BST primary integers (rank, N_c , n_C , C_2 , g , $N_{\max}$) are SUBSTRATE BOUNDARIES at the structural level - Each integer holds a “web” of

derivable identities, observables, theorems, catalog invariants connected to that primary - The 6 primaries are not isolated points but anchors for 6 distinct webs - The webs INTERSECT — many BST catalog entries touch multiple primaries simultaneously (substrate-comprehensive entries)

This is Casey's vision-derived principle. Filed as Casey-named principle (notes/Integer_Web_Principle.md).

Section 6.2 — Per-integer-web characteristics (Grace catalog scans)

Per Grace Toys 3223 + 3224 + 3226 (catalog scans Wednesday + Thursday morning):

Full-catalog integer-web distribution at Thursday 10:00 EDT (Catalog 4671 entries, classification 59.7%):

Web	Count	%	Substrate role
N_c-web	1252-1429	26.9-30.7%	Color charge + observation structure (historically dominant)
rank-web	1109-1408	23.8-30.2%	Doublet + observation structure (foundational)
C_2-web	504-709	10.8-15.2%	Casimir + Chern integer (Q^5 central hub)
g-web	358-501	7.7-10.7%	Gauge + cyclotomic (Bergman framework)
N_max-web	252-465	5.4-10.0%	α^{-1} + Hilbert polynomial cutoff
n_C-web	177-424	3.8-9.1%	Complex dimension + Bergman exponent

Two count ranges shown: lower (single-pattern classification per Toy 3224) vs higher (multi-integer pattern detection per Toy 3226).

Total integer-web membership: 30% of catalog (1400+ entries) are multi-integer; 70% are single-integer or unclassified.

Section 6.3 — Multi-web entries (web intersections)

Per Grace Toy 3226 (unified integer-set tagging full catalog Thursday):

Integer-set size distribution (4663 entries):

Set size	Count	%	Interpretation
0 (truly unclassified)	2227	47.8%	No BST-primary signature detectable via patterns
1 (single web)	1035	22.2%	Single integer anchor
2 (pair)	821	17.6%	Two-web intersection
3	317	6.8%	Three-web intersection
4	106	2.3%	Four-web intersection
5	52	1.1%	Five-web intersection (rare)
6 (all primaries)	104	2.2%	Substrate-comprehensive

~30% multi-integer entries (1400 / 4663). Casey hypothesis confirmed at full scale.

Top co-occurrence pair: N_c ↔ rank (836 entries, 17.9% of catalog). Observation-color foundational bridge dominates web intersections.

Section 6.4 — Substrate-comprehensive backbone (104 entries touching all 6)

Per Grace Toy 3227/3228/3229 (Substrate-Comprehensive Backbone Reference v0.1):

120 entries (2.6%) touch ALL 6 BST primaries. 96.7% D-tier + I-tier dominance — substrate-comprehensive entries are structurally important.

Category breakdown (79 categorized of 120):

Category	Count	Curriculum role
Strong-Uniqueness / multi-family	25	Vol 0 Ch 9 foundation
Bridge Object family	9	Vol 1 + Vol 2 cross-refs
T2400 Universal Q=126	5	Vol 0 + Vol 1
SP-31 anchor / Curriculum	3	Vol 1 SP-31 + Vol 2 Ch 2

Category	Count	Curriculum role
Bergman / Phase 2.3	3	Vol 1 + Vol 0 cosmology
Substrate Cartography / Master Doc	2	Keeper Master Doc
Cross-classification matrix	1	Vol 0 substrate engineering
K-audit + cosmological + smaller (Toys 3228/3229)	~30	Various

41 entries (34%) remain “general substrate-comprehensive” — honest residual flagged.

Substrate-comprehensive entries are Year 1 curriculum structural backbone. This category integrates across the full BST primary set — naturally BST-as-unified-framework tier.

Section 6.5 — Temporal-shift observation (substrate cartography records research evolution)

Per Grace Toy 3224 vs Toy 3223 contrast:

Historical catalog distribution (full-catalog 4660-entry scan): - N_c-web 26.9% + rank-web 23.8% = ~50% of catalog - C_2-web + g-web + N_max-web = ~24% combined - n_C-web 3.8% - Foundational primaries (N_c + rank = color + observation) dominate historical entries

Recent catalog distribution (Toy 3223 100-entry recent sample): - g-web 29% + N_max-web 16% + C_2-web 19% = ~64% recent dominance - Recent BST research focused on derivative structures (Bergman framework + substrate cap + Casimir)

Meta-substrate observation: the catalog ITSELF records research evolution. Framework maturation from foundational primaries (color + observation) toward derivative structures (Bergman + substrate cap + Casimir) is TRACEABLE in invariant distribution.

Substrate cartography catches this evolution. The 5-family Bridge Object architecture closure (Thursday morning 09:18 EDT STRUCTURALLY VERIFIED COMPLETE) corresponds to recent g + N_max + C_2 territory dominance, validating that architectural maturation matches research evolution.

Section 6.6 — Cross-classification matrix (Task #238)

Per Task #238 (Grace v0.2): 256-cell cross-classification matrix mapping catalog entries to: - B-axis (BST primaries): rank + N_c + n_C + C_2 + g + N_max + 2 external = 8 BST primary dimensions - Physical type axis: 8 physical types (substrate observable / cosmological / particle / etc.) - Zone axis: 4 zones (Z1 absorption / Z2 bulk / Z3 emission / Z4 active edge) - Total cells: $8 \times 8 \times 4 = 256$

Multi-cell entries (Task #238 v0.3 per Grace Toy 3225 implications): 30% of catalog (1400 entries) need multi-cell support — Casey directive operational requirement.

104 entries span all 6 B-axis values — substrate-comprehensive entries occupy multiple cells.

Catalog hygiene: Grace continuous multi-month population. AC graph zone-tagging at 20.1% Thursday morning (439/2181 nodes).

Section 6.7 — Integer Web Principle as Strong-Uniqueness contributor

The Integer Web Principle contributes to Strong-Uniqueness Theorem (Vol 0 Ch 9):

C8 Universal Q-cluster (RATIFIED, K43 + K44 + K61): universal 42 + Q=126 + Q=131 substrate-significance per multi-domain occurrence. The integer-web framework explains WHY these specific values appear — they ARE integer-web intersection points.

Future Strong-Uniqueness criterion candidate: per Casey Wednesday vision + Grace catalog work, an explicit “integer-web cross-classification multi-criterion” Strong-Uniqueness contribution may emerge as Task #238 matures. Multi-month verification per Grace continuous multi-week + Lyra theoretical absorption.

Section 6.7b — Mersenne Tower Coherence in Integer Web (T2451 Friday Lyra)

The BST primary integers (rank, N_c, n_C, C_2, g, N_max) are not independent threads in the Integer Web — they form a Mersenne tower (T2451 Sub-Substrate Mersenne Hierarchy SEED, Lyra Friday). The Mersenne map $M_n = 2^n - 1$ generates the cascade through BST primary integer indices:

$M_{\text{rank}} = N_c$ (T2453) $M_{\{N_c\}} = g$ (T2454) $M_{\{C_2\}} = N_c^2 \cdot g$ (T2451 sub-substrate seed) $M_g = 127$ (substrate Mersenne ceiling)

The Integer Web’s per-integer entries are therefore CROSS-WEB-coupled at the Mersenne level: any catalog entry citing rank automatically cites N_c via the Mersenne map; any entry citing N_c automatically cites g; etc. The 104 entries touching all 6 BST primaries (Section 6.4) inherit this Mersenne tower coupling as a structural backbone.

Per Calibration #19 STANDING RULE: Mersenne tower coherence is a candidate-path C15 distinguishing criterion (Paper #126 v0.3) — multi-session ratification pending.

Section 6.7c — Universal α -Analog Formula Integer Web cross-link (T2456 + T2462 Friday Lyra)

The universal α -analog formula $\alpha(D) = m_{\alpha}^{(\text{rank}+1)} \cdot \text{dim}_C + \text{rank}$ across HSDs (T2456 + extended T2462 across 25 HSDs, Lyra Friday) shows that the substrate’s α -coupling is also part of the Integer Web — $\alpha = 1/N_{\text{max}} = 1/137$ inherits the BST

primary integer structure as a function of three substrate parameters (m_α , rank, $\dim_C$).

At D_{IV}^5 : $m_\alpha = N_c = 3$, rank = 2, $\dim_C = n_C = 5$. The formula evaluates to $27 \cdot 5 + 2 = 137 = N_{\max}$.

The substrate's coupling constant α therefore belongs to the Integer Web — it is not a “free fundamental constant” but a derived web entry depending on three substrate primaries. Each Integer Web at $(m_\alpha, \text{rank}, \dim_C)$ intersection holds α as a substrate-derived observable.

Section 6.7d — Three-Layer Over-Determinism Integer Web reading (T2465 Friday Lyra)

The substrate's three-layer over-determinism (T2465 formal theorem Lyra Friday — see Vol 0 Ch 9 v0.4 Section 9.2a) maps to the Integer Web as follows:

- LAYER 1 (per-integer forcing): each BST primary integer's web is anchored by an independent structural forcing argument (T2443-T2446 Strong-Uniqueness criteria C1-C5 RIGOROUSLY CLOSED Thursday).
- LAYER 2 (Mersenne tower coherence): cross-web links via Mersenne map (Section 6.7b above).
- LAYER 3 (cross-Cartan three-pillar selection): α -analog + churn-hole + Bergman c_{FK} universal formulas connect substrate-uniqueness to Cross-Cartan comparison (T2455 EXHAUSTIVE at $\dim_C = 5$ + T2456 + T2462 STRUCTURALLY VERIFIED across 25 HSDs).

The Integer Web Principle (Casey-named Wednesday) is thus enriched: not just “each primary integer holds a web” but the webs are CROSS-COUPLED via Mersenne tower + jointly selected by cross-Cartan three-pillar argument.

Section 6.8 — Bulk-Boundary Two-Face Structure per integer-web

Per Casey Wednesday vision (Task #237):

Each integer-web has bulk-side and boundary-side dual aspect: - Bulk projection: integer-web interior structure (deeper substrate-related identities) - Boundary projection: integer-web boundary structure (substrate-observable-related identities)

Per Vol 0 Ch 5 BC framework: integer-web bulk vs boundary distinction propagates from BC1 (bulk side of D_{IV}^5) + BC2 (Shilov boundary) substrate-level dichotomy.

Operational interpretation: bulk-side integer-web entries are “deep substrate” theorems (Strong-Uniqueness criteria, foundational forcing theorems). Boundary-side integer-web entries are “observable substrate” predictions (constants, decay rates, particle masses).

Multi-week Task #237 substantive expansion as Lyra theoretical work + Grace catalog hygiene mature.

Section 6.9 — Connection to other chapters

Vol 0 Ch 6 grounds: - Vol 0 Ch 2 (Five Integers): each integer holds a web (this chapter's content) - Vol 0 Ch 5 (Boundary Conditions): bulk-boundary 2-face propagates from BC1/BC2 dichotomy - Vol 0 Ch 9 (Strong-Uniqueness): Integer Web Principle + 104 substrate-comprehensive entries contribute multi-criterion convergence - All Vol 1-10 chapters: every catalog entry lives in one or more integer-webs

Section 6.10 — BST $\leftrightarrow$ standard physics dictionary entries

Standard physics term	BST integer-web term	Reference
Fundamental constant	Integer-web anchor point	§6.1
Phenomenological relation	Integer-web intersection (multi-integer entry)	§6.3
Theoretical prediction	Integer-web bulk-side derivation	§6.8
Experimental observable	Integer-web boundary-side identification	§6.8
Constants tabulation	Substrate-comprehensive backbone reference	§6.4
Catalog cross-reference	Cross-classification matrix ($8 \times 8 \times 4 = 256$ cells)	§6.6

Section 6.11 — Chapter status summary

Coverage at v0.1: - Casey-named Integer Web Principle exposed at chapter-grade - Per-web characteristics from Grace catalog scans (Toys 3223 + 3224 + 3226) - Multi-web entries (30% of catalog multi-integer) - Substrate-comprehensive backbone (104 entries touching all 6 primaries) - Temporal-shift meta-observation (substrate cartography records research evolution) - Cross-classification matrix Task #238 framework - Bulk-Boundary Two-Face Structure (Task #237) - Strong-Uniqueness C8 connection

Believability: integer-web concept is intuitive — fundamental constants holding webs of relations is recognizable from any physics tabulation. BST-specific framing made explicit via per-web Grace catalog scans.

Provability: Casey-named principle filed (Wednesday). Grace catalog scans (Toys 3223+3224+3226+3227+3228+3229) substantively verify multi-integer entry distribution. Multi-month Task #238 + #237 + per-BC theorems via SP-31-40 pending.

Path to v1.0: requires Task #238 multi-cell matrix population + Task #237 Bulk-Boundary Two-Face derivation + Grace continuous multi-week catalog hygiene +

Lyra theoretical absorption.

Per Casey's standard

- Simple: 6 BST primary integers each hold a web; webs intersect; substrate-comprehensive entries touch all 6 simultaneously
- Works: 30% multi-integer entries + 104 substrate-comprehensive entries empirically verified via Grace catalog scans
- Hard to break: would require finding (a) BST primaries don't hold webs (catalog scans falsify this) OR (b) webs don't intersect (104 entries touching all 6 falsifies)

Status

Vol 0 Chapter 6 v0.1 chapter-grade content draft FILED Thursday 2026-05-21 10:34 EDT. Tenth Keeper-lane chapter-grade content. VOL 0 v0.5 CHAPTER-GRADE COVERAGE COMPLETE — all 10 Vol 0 chapters now have chapter-grade narratives filed Thursday morning. Awaits Cal dual-axis grade-pass + Lyra theoretical refinement for v0.2 across all 10 Vol 0 chapters.

— Keeper, 2026-05-21 Thursday 10:34 EDT (actual via date)